Analysis of Job Market Data

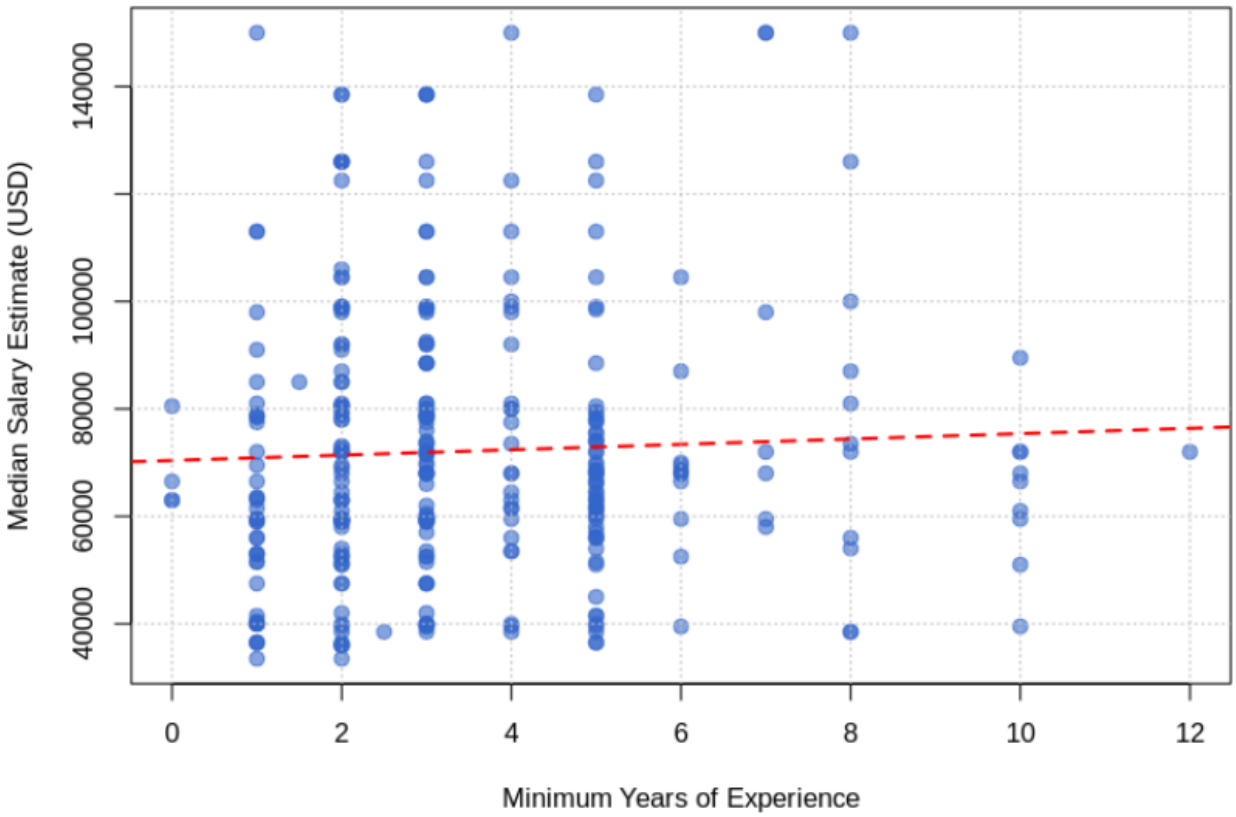
Executive Summary

This report analyses job market data for data analyst positions using Large Language Models (LLMs) to extract specific requirements from job descriptions. The analysis focuses on two key aspects: experience requirements and programming language preferences, providing insights into current market demands and salary relationships.

Methodology Note

This analysis was conducted using a multi-step approach combining LLMs for data extraction with traditional statistical analysis tools. Job descriptions were processed using Google/Gemini-2.5-flash model through OpenRouter API integration in Google Sheets, followed by visualization and analysis using Julius AI agent with Claude 4 Sonnet model.

Minimum Years of Experience vs. Median Salary Estimate



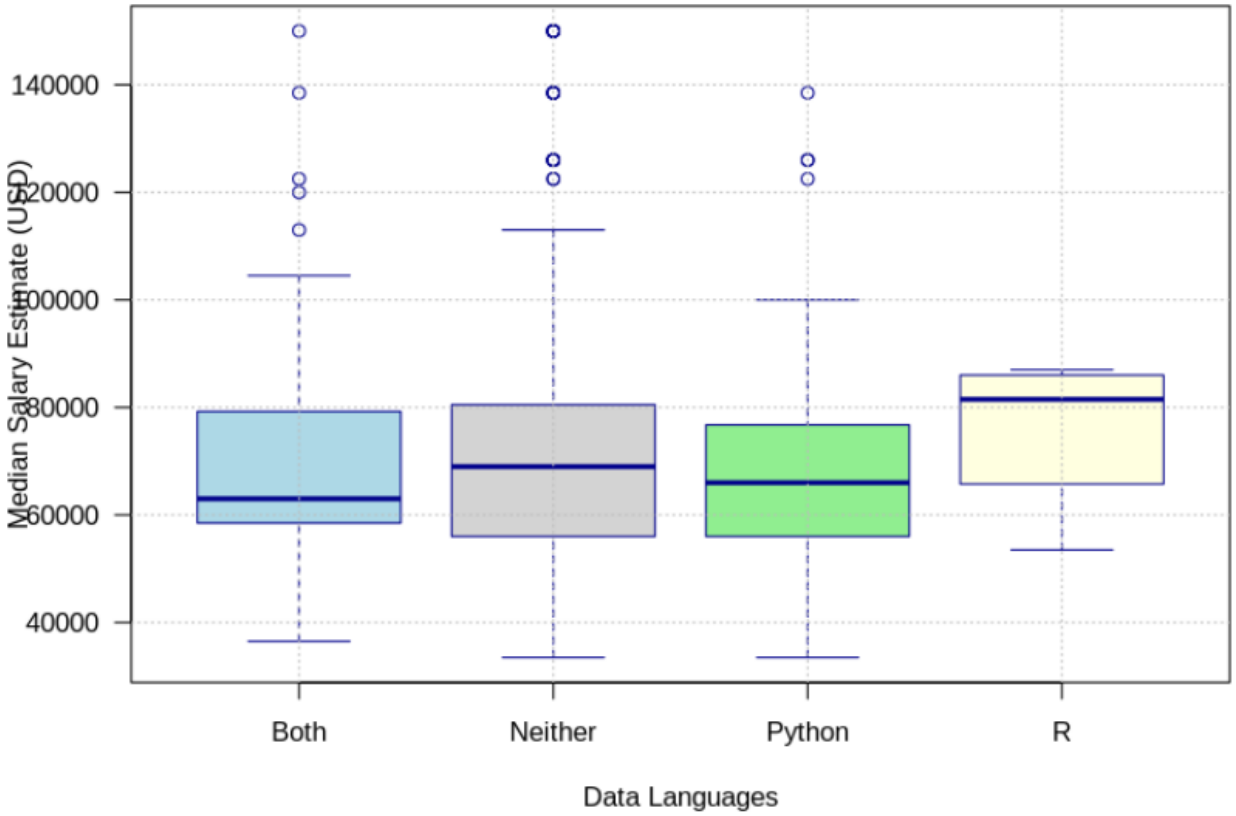
Key Findings

This scatter plot reveals the relationship between minimum years of experience required and median salary estimates in data analyst positions. The analysis shows a positive correlation between experience requirements and compensation, as indicated by the red dashed trend line. Notably, 310 out of 400 job postings contained valid numeric experience values, suggesting that most employers specify experience requirements explicitly.

The data demonstrates that entry-level positions (0-2 year's experience) show considerable salary variation, ranging from approximately $40,000 to over $100,000, indicating diverse market segments and geographic variations. As experience requirements increase, salaries generally trend upward, though with notable scatter

around the trend line. Positions requiring 5+ years of experience consistently show higher median salaries, typically above $70,000.

Median Salary Estimate by Data Language



Key Findings

This box plot compares salary distributions across different programming language requirements, revealing interesting patterns in the data analyst job market:

- Both (R and Python): 56 jobs - Shows the highest median salaries and narrowest interquartile range, suggesting that dual-language proficiency commands premium compensation with less variability.
- Neither: 293 jobs - Represents the largest category, indicating many data analyst positions don't require specific programming languages. This category shows wide salary variation.
- Python only: 47 jobs - Demonstrates competitive salaries with moderate variation, reflecting Python's popularity in data science workflows.
- R only: 4 jobs - The smallest category, making statistical conclusions limited, but suggests specialized roles with competitive compensation.

The analysis indicates that programming language requirements significantly influence compensation, with positions requiring both R and Python offering the most competitive and consistent salaries. This suggests that versatility in programming languages is highly valued by employers.

Personal Reflection on the Process

The task was easier than I expected, mainly because the instructions were very clear and easy to follow. Even though I do not have a background in statistical analysis, the objective of the exercise was straightforward, and the results produced by Julius were clearly presented and easy to understand.

What I found most insightful was seeing how these tools can be used not only to clean or collect data, but also to extract information from a large number of entries and turn it into new comparisons and analyses.

It was also interesting to learn that this kind of data analysis can be done directly in Google Sheets using specialized online tools. These tools make it possible to carry out tasks that would normally require dedicated software or programming skills.

While the exercise relied heavily on formula language, which I am not very familiar with, the availability of good documentation and AI support makes it much easier to work with. Overall, it showed how these tools can help improve workflows and make both regular and occasional data tasks more manageable.